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Communicating climate science: The role of perceived communicator's motives

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ABSTRACT

In two experimental studies, we investigated the effects of public perceptions of climate scientists' communicative motives on trust in scientists and willingness to engage with climate science messages. Study 1 demonstrated that members of the public who were led to believe that scientists aim to inform about the consequences of climate change (rather than to persuade to take a particular course of action) reported higher trust in scientists and stronger willingness to engage in environmental behaviour. Study 2 revealed that this effect was moderated by the style of the scientific message that participants were exposed to. Participants who expected scientists to engage in persuasion were more receptive to persuasive rather than informative messages, while the opposite was true for participants who believed that scientists' purpose was purely to inform. In both studies the effects of perceived motives on willingness to act in line with the climate change messages were mediated through trust in scientists. The data demonstrate that managing public expectations about the purposes of science communication and delivering messages that are consistent with these expectations are a key to successful communication of climate science.

1. Introduction

Misunderstandings between the scientific world and the wider public are a recognisable phenomenon within contemporary society. With the increasing speed of technological and social change, the increasing visibility of scientists as communicators of their work, and a growing audience for scientific information, examples of misunderstanding are also becoming more numerous (including DNA research, genetically modified food, and notably, climate change). Because scientific messages are becoming ever more complex, fostering effective communication between the scientific world and the general public is more challenging, but also more important than ever before. In this setting, social psychological theory and research may offer useful insights about the means and mechanisms through which more effective communication might be achieved.

Much previous research on the psychology of communicating climate science has concentrated on properties of the message and how these might influence message understanding and acceptance (e.g., Budescu, Broomell, & Por, 2009; Gifford & Comeau, 2011; Morton, Rabinovich, Marshall, & Bretschneider, 2011). However, more communicatively oriented research would suggest that the qualities of a message may not be the only, or even the most decisive, aspect of the communication process. In fact, the success of communication often does not reside within the message itself (however masterfully it may be constructed and framed), but within the dynamic interaction between the communicating partners (cf. Delia, O'Keefe, & O'Keefe, 1982; see also Mackie, Worth, & Asuncion, 1990). Of paramount importance to such interactions is communication partners' ability to recognise each other's position and the fact that this position may be different from one's own (e.g. Fussell & Krauss, 1992; Wilkes-Gibbs & Clark, 1992). One consequence of this ability is understanding that one's partner may have distinct motives for communication and accommodating these motives to facilitate exchange and mutual influence (see Giles, Coupland, & Coupland, 1991). The role of these interactive processes has been underexplored in the domain of science communication. The present paper aims to address this gap by exploring how message recipients' beliefs about communicators' motives affect the process of climate change communication.

We start by briefly reviewing psychological research on the role that communication partners' beliefs about each other play in the process of communication. Based on this research, we derive hypotheses about the effects of perceived communicator's motives on effectiveness of science communication, and then test these hypotheses in two experimental studies.

1.1. Beliefs, expectations, and effective communication

The idea that communication partners' beliefs about each other's qualities and motives might affect the process of communication

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has been widely explored in psychology. For example, it has been demonstrated that an ability to recognise agency in one's communication partner (so called "mind perception", see Gray, Gray, & Wegner, 2007), as well as to interpret and predict one's partner's moves, is a crucial pre-condition for successful communication (e.g., Keysar, Barr, Balin, & Brauner, 2000; Schober, 1993) and something that seems to be unique to human beings (see Hare, 2007; Herrman, Call, Hernandez-Lloreda, Hare, & Tomasello, 2007).

Not only are people capable of recognising others' knowledge and motives in communication process (e.g., Krauss & Fussell, 1996), but they also seem to be skilful in adapting their own responses to what they believe about their interaction partner (i.e., communication accommodation: Giles et al., 1991; see also Hardin & Higgins, 1996; Vogel, Kutzner, Fiedler, & Freytag, 2010). For example, communicators may shift content and style of their messages either towards what they believe to be the recipient's position (i.e., convergence) or away from it (divergence). The choice of one of these strategies depends on communicator's beliefs about recipient's status and likeability – while the convergence strategy may be used to gain approval of a likeable or a high status person, the latter strategy (divergence) may be used to maintain psychological distance and authority over a recipient with a lower status (Giles et al., 2002). Finally, the process of accommodating (or diverging) has consequences for the outcomes of communication. For example, through accommodating to a communication partner's expectations, people may end up confirming their partner's stereotypes about their self, or the group to which they belong (see Klein & Snyder, 2003). Moreover, accommodating (or diverging) during communication has consequences for one's subsequent beliefs as people tend to remember what was said during an exchange (which may be a product of accommodation) rather than what they initially believed prior to communication (i.e., the "saying is believing" effect: Echterhoff, Higgins, & Groll, 2005; Echterhoff, Lang, Kramer, & Higgins, 2009).

Of course, recipients also do not stay passive in the process of communication. According to the persuasion knowledge model (Friestad & Wright, 1994), recipients are motivated to develop valid perceptions of communicators (e.g., persuasive agents) and to hold valid attitudes towards them. Indeed, there is empirical evidence that people are capable of making accurate inferences about their interaction partners, even when only minimal information is available (Ambady, Krabbenhoft, & Hogan, 2006; Friestad & Wright, 1995). Message recipients may also use this knowledge strategically by adjusting their interpretation of the message in line with what they believe about communicators. In extreme cases, recipients may engage in active rejection of the message – for example, when they believe that a communicator has a hidden agenda (see Jacks & Devine, 2000; Knowles & Linn, 2004; Petty & Cacioppo, 1977, 1979). Beliefs about communicators' motives affect recipients' interpretation of the message not only directly (e.g., through stimulating active rejection in response to insincerity), but also indirectly through the process of trust. Previous research demonstrated that recipients sometimes anticipate specific biases from a communicator (e.g., based on the latter's background and experience), and that the extent to which the message is consistent with expected biases has consequences for trust (Eagly, Wood, & Chaiken, 1981). Somewhat paradoxically, communicators are trusted more when the position they advocate is *inconsistent* with the bias that was initially expected from them (Eagly, Wood, & Chaiken, 1978; Peters, Covello, & McCallum, 1997; Wood & Eagly, 1981). For example, when a green party member advocates against the introduction of a green tax, she may be seen as particularly persuasive and trustworthy. The reason is that when people are faced with a message that contradicts the expected communicator's bias, they look for an alternative motive behind the message. One of the readily available explanations is that the facts must be so compelling that they led the communicator to abandon their typical biases (see Wood & Eagly, 1981). Thus beliefs about communicators can frame an audience's interpretation of the positions they advocate: if the advocated position is inconsistent with the expectations, it may come to be seen as strongly justified. In addition, disconfirming expectations can also have implications for evaluations of the communicator: speakers who deliver such disconfirming messages are also seen as more trustworthy (Eagly, Wood, & Chaiken, 1981, see also Bohner, Erb, Reinhard, & Frank, 1996; Bohner, Frank, & Erb, 1998; Moskowitz, 1996).

1.2. Beliefs and expectations in science communication

Research into the role of beliefs and expectations about one's communication partner in framing both message production and reception has direct implications for science communication. Although some research in this domain has concentrated on the role of scientific arguments *per se* (e.g., Corner & Hahn, 2009; Morton et al., 2011), it is clear that scientific arguments are not considered in isolation from those who communicate them. Quite obviously, science communicators (e.g., climate scientists) and general public have expectations and beliefs concerning each other (e.g., Davies, 2008; Frewer et al., 2003; Locke, 1999; Terwel, Harinck, Ellemers, & Daamen, 2009). Far from being passive recipients of scientific information, various sections of the public have been found to hold strong preferences and expectations about science and the ways in which it is communicated (e.g. Horst, 2007; Vandermoere, Blanchemanche, Bieberstein, Marette, & Roosen, 2011). On the basis of the research discussed above, it is reasonable to suggest that such public expectations should play a significant role in determining outcomes of science communication (cf. Eagly, Wood, & Chaiken, 1981).

Indeed, previous research in the domain of science communication suggests that perceived communicators' motives play an important role in determining trust in communicators (e.g., Ter Mors, Weenig, Ellemers, & Daamen, 2010; Terwel et al., 2009). For example, Terwel et al. (2009) demonstrated that when organisations communicated motives inconsistent with those expected from them (e.g., industrial organisations communicated public-serving motives, or non-governmental environmental organisations communicated organisation-serving motives), this undermined trust in such organisations and their messages. Interestingly, these findings contradict previous research by Eagly and colleagues (e.g., Eagly, Wood, & Chaiken, 1981), demonstrating that sometimes consistency between expected and communicated motives is more conducive to trust development than perceived ability to overcome one's biases (i.e. a "known evil" is preferred to an "unknown" one).

Before further exploring the issue of consistency between recipients' expectations and perceptions in science communication, it is first important to address the question of how specific science communicators' motives affect recipient responses. Based on past research, it could be expected that recipients of scientific messages who suspect a hidden agenda behind the message will attempt to resist persuasion (cf. Jacks & Devine, 2000; Knowles & Linn, 2004; Ter Mors et al., 2010). One specific form of hidden agenda that is often raised in public debates about science is that scientists have "an axe to grind" and that they are stepping beyond their expected role of information providers by engaging in influence and persuasion. Ter Mors et al. (2010) provided some indirect evidence that self-serving motives perceived behind science communication are detrimental to trust. Specifically, they showed that sets of diverse collaborating organisations incite stronger trust in the scientific information they produce than individual organisations – presumably, because recipients assume that collaborative units are less likely to pursue self-serving motives. Thus, the perception that scientists are attempting to manipulate or persuade, rather than to inform, may result in decreased trust in scientists and reduced willingness to engage with communications from this source (e.g., by accepting their message). In contrast, the perception that scientists do not have a hidden agenda, and are merely conduits of information rather than influence, may facilitate trust and message acceptance. We start by testing this basic hypothesis in Study 1 in the context of climate change communication.

2. Study 1

2.1. Method

2.1.1. Participants and design

Fifty-seven adults participated in the study (14 males and 43 females, mean age 28 years). Participants were randomly recruited in public spaces of a British city. A between-subjects design was used with two experimental conditions in which we attempted to vary the image of scientists' communicative agenda (i.e., to inform vs. to persuade). Participants were randomly allocated to one of the two conditions. The dependent variables were participants' trust in scientists and intention to engage in environmentally sustainable behaviour.

2.1.2. Materials and procedure

Participants were approached individually and asked to complete a short questionnaire. On the first page, participants read an excerpt from a (fake) newspaper article describing a survey conducted among climate scientists. The content of this article was varied to manipulate perceptions of climate scientists' motives in communication. Participants in one condition read that 80% of climate scientists claimed that they aim solely to inform the public and decision-making bodies about the consequences of climate change, to produce precise and impartial information, and that they did not have hidden agenda. Thus scientists were portrayed as being motivated *to inform*. In the alternative condition, participants instead read that 80% of climate scientists reported that they aim to persuade the public and decision-making bodies to take action on climate change, promote belief in human-induced climate change and stimulate action. Thus in this condition, scientists were portrayed as being motivated *to persuade*. To verify that the manipulation was successful, participants were asked to indicate if they agreed or disagreed with the two statements "most climate scientists believe that their aim is to provide impartial information" and "most climate scientists believe that their aim is to persuade the public to take action" (on a 7-point scale from 1 "strongly disagree" to 7 "strongly agree").

2.1.3. Dependent measures

After the manipulation check, participants completed measures of trust in climate scientists and intentions to engage in environmentally sustainable behaviours (i.e., personal actions relevant to the communicators' goals).

Two items were used to measure participants level of trust in climate scientists: "I trust climate scientists as a source of information" and "I do not trust climate science" (recoded), ($r\,(57) = .528$, $p < .001$). Participants responded to these items on a 7-point scale from 1 "strongly disagree" to 7 "strongly agree." The two items were averaged to from a single measure of trust.

To measure environmental intentions, participants were asked to indicate how likely they were to engage in a number of environmentally sustainable activities over the following month (e.g., "reduce water use" and "join in community environmental activities", 10 items, alpha = .87). Participants responded to these items on a 7-point scale from 1 "very unlikely" to 7 "very likely." After completing these measures and answering some demographic questions, participants were thanked and debriefed.

2.2. Results

2.2.1. Manipulation check

The analysis of the manipulation check items demonstrated that the manipulation of scientists' purpose was successful. Participants in the "purpose to inform" condition were more likely to agree that scientists' purpose is to inform ($M = 5.76$, $SD = .99$) than participants in the "purpose to persuade" condition ($M = 4.25$, $SD = 1.53$; $t\,(45.92) = 4.41$, $p < .001$, $d = 1.17$). Similarly, participants in the "purpose to persuade" condition were more likely to agree that scientists' purpose is to persuade the public to take action ($M = 5.46$, $SD = 1.82$) than participants in the "purpose to inform" condition ($M = 3.41$, $SD = 1.74$; $t\,(55) = 4.35$, $p < .001$, $d = 1.15$).

2.2.2. Main analysis

To explore the effect of our manipulation on participants' trust in scientists and willingness to engage in sustainable activities, we conducted two independent samples *t*-tests. Participants in the "inform" condition reported higher trust in climate scientists ($M = 5.34$, $SD = 1.08$) than participants in the "persuade" condition ($M = 4.79$, $SD = 1.05$, $t\,(55) = 1.98$, $p = .05$, $d = .52$). Participants in the "inform" condition also reported stronger environmental intentions ($M = 3.43$, $SD = 1.34$) than participants in the "persuade" condition ($M = 2.73$, $SD = .85$; $t\,(47.65) = 2.37$, $p = .022$, $d = .62$).

2.2.3. Mediation analysis

Given these parallel effects on trust and action, and given the often theorized role of trust as the basis for communication effects, we tested whether the effect of the manipulation of perceived scientists' purpose on willingness to act sustainably was mediated through trust. Mediation analysis was performed via regression. This analysis revealed that perceived purpose to persuade had a significant negative effect on trust, $\beta = -.26$, $t\,(55) = -1.98$, $p = .05$, and a significant negative effect on sustainable intentions, $\beta = -.30$, $t\,(55) = -2.35$, $p = .022$. However, when both experimental condition (perceived purpose) and trust were included as predictors of environmental intentions, the effect of experimental condition on intentions was no longer significant ($\beta = -.22$, $t\,(55) = -1.72$, $p = .092$), and trust in scientists was the only significant predictor of intentions to act, $\beta = .33$, $t\,(55) = 2.62$, $p = .011$. The bias corrected bootstrap estimate of the indirect effect had a 95% confidence interval of $-.5816$ to $-.0316$. As this confidence interval does not span zero, this suggests that the effect of perceived scientists' communicative purpose on intentions to engage in sustainable behaviours was mediated by trust in climate scientists (see Preacher & Hayes, 2004).

2.3. Discussion

Study 1 provided support for our basic hypothesis that perception of a hidden agenda in science communication, specifically an agenda to engage in persuasion rather than the provision of information, can result in reduced trust in scientists and reduced willingness to engage with the issues being communicated (cf. Jacks & Devine, 2000). While this suggests that perceived motives might impinge on public receptivity to science communication, it is important to note that participants in Study 1 were not exposed to any communicative messages from climate scientists. Thus, while our data show that the perception of a persuasive agenda might reduce engagement with issues of concern to the messenger, it is unclear how such perception would interact with the message that is ultimately delivered.

3. Study 2

To address the limitations of the first study, in a second study we explored how perceived motives influenced responses to different types of messages, specifically messages that were constructed in an informative versus persuasive style. Thus, in this study, we ask what happens when an audience's expectations about the motives of a communicator are confirmed or disconfirmed by the message they ultimately deliver.

Based on research by Eagly and colleagues (Eagly, Wood, & Chaiken, 1981), it could be expected that people who perceive the communicator's goals as being to persuade but instead receive a neutral and informative message, should experience elevated levels of trust in the communicator and be more willing to act in line with their message (i.e., due to the assumption that the communicator has overcome his or her bias and stuck to the facts). In contrast, people whose expectation of a persuasive agenda is confirmed by the delivery of a persuasive message should report reduced trust and become motivated to resist the communicator (cf. Jacks & Devine, 2000; Petty & Cacioppo, 1979).

Although these predictions could be made based on previous research in other domains, it also seems plausible that people may not respond so straightforwardly to disconfirmed expectations in the domain of science communication. Specifically, people who expect that a communicator has a persuasive agenda may become vigilant against any signs of persuasion, even when they are presented with a purely informative message. This level of suspicion may interfere with receptiveness to messages of any kind. Similarly, people who expect scientists to be neutral and to engage in purely informative (rather than persuasive) communication, may become sceptical when these expectations are violated by a persuasive message (cf. Jacks & Devine, 2000). In line with this reasoning, Terwel et al. (2009) found that messages inconsistent with motives attributed to a particular organisation have a detrimental effect on trust.

This alternative line of thinking would lead to the prediction that disconfirmation of expectancies would be particularly damaging for trust and engagement from the audience. Conversely, fit between public's expectations about the purpose of scientific communication and messages that confirm these expectations may result in increased trust and persuasiveness. However, the beneficial effects of confirmed expectations, and the deleterious effects of disconfirmation, might not hold in the context of a persuasive agenda. Here, the distrust aroused by the expectation of persuasion might lead to scepticism and colour the perception of messages of any kind, regardless of whether they are neutral or persuasive. To explore these various possibilities, we manipulated perceived scientists' communicative purpose (informative vs. persuasive) and message style (informative vs. persuasive) orthogonally, and tested the interactive effect of these parameters on trust and willingness to engage with climate change issues.

3.1. Method

3.1.1. Participants and design

Participants were students and staff members of a British University recruited via email. One hundred and fourteen people (37 male, 77 female, mean age 26 years) completed an online questionnaire in exchange for being entered into a prize draw. A 2×2 factorial design was used with perceived scientists' purpose (to inform versus to persuade) and message style (informative versus persuasive) as independent variables. Participants were randomly assigned to each of the four experimental conditions. Dependent variables were trust in climate scientists, environmental concern, and participants' intentions to engage in environmentally sustainable practices.

3.1.2. Materials and procedure

Participants completed the questionnaire online. Perceived scientists' purpose was manipulated and checked in the same way as in Study 1. On the following page, the type of climate change message was manipulated. Participants were asked to read a short text about the consequences of climate change; they were led to believe that the text was authored by climate scientists. Participants in the "informative style" condition read a bullet point list of facts about climate change titled "Facts about Climate Change" (e.g., "over 40% of current CO_2 emissions are caused by the choices we make as individuals"). Participants in the "persuasive style" condition read a text entitled "Fact and Fiction about Climate Change." This text consisted of the same list of facts as in the "informative style" condition, but each fact was preceded by a "fiction" statement that the following "fact" statement proved wrong (e.g., "Fiction: My small choices don't make any difference. Fact: Over 40% of current CO_2 emissions are caused by the choices we make as individuals"). Thus while the "informative" text consisted of facts only, the "persuasive" text was constructed in a rhetorical style suggesting that there are certain beliefs that need to be disproved. To check if the manipulation of message style was successful, participants were asked to indicate if they agreed or disagreed with the following statement: "the purpose of this article was to persuade people to take action on climate change" (on a 7-point scale from 1 strongly disagree to 7 strongly agree).

3.1.3. Dependent measures

After completing the manipulation check, participants completed measures of trust in climate scientists, environmental concern, and intention to engage in environmentally sustainable behaviour. Unless otherwise stated, participants responded to all items on a 7-point scale from 1 (strongly disagree) to 7 (strongly agree).

To measure trust, five items were used, including two items used in Study 1(e.g., "If I hear a statement coming from a climate scientist I generally believe it is true", alpha = .84). Two items were used to measure environmental concern: "I am very concerned about the effects of climate change" and "The consequences of climate change may be extremely serious" ($r\,(113) = .65$, $p < .001$).

To measure environmental intentions, a subset of the items from Study 1 was used (eight items, alpha = .84). Participants were asked how likely they were to perform a number of environmentally sustainable actions during the next months (e.g., "wash clothes at below 40°" and "donate money to environmental charities") and responded on a 7-point scale from 1 (very unlikely) to 7 (very likely). After completing the questionnaire, participants were thanked and directed to a debriefing page.

3.2. Results

3.2.1. Manipulation checks

The manipulation of participants' perception of climate scientists' communicative purpose was successful. Participants in the "purpose to inform" condition were more likely to agree that scientists' purpose is to inform ($M = 5.57$, $SD = 1.42$) than participants in the "purpose to persuade" condition ($M = 2.86$, $SD = 1.63$; $t\,(110) = 9.32$, $p < .001$, $d = 1.77$). Similarly, participants in the "purpose to persuade" condition were more likely to agree that scientists' purpose is to persuade the public to take action ($M = 5.86$, $SD = 1.16$) than participants in the "purpose to inform" condition ($M = 2.94$, $SD = 1.85$; $t\,(85.89) = 9.83$, $p < .001$, $d = 1.89$).

To check whether the manipulation of message style was successful, we conducted a 2 (message style: informative vs. persuasive) $\times$ 2 (perceived purpose: to inform vs. to persuade) ANOVA on the relevant manipulation check item. The only significant effect was that of the message style condition: participants in the "persuasive style" condition were more likely to agree that the purpose of the text was to persuade people to take action ($M = 6.12$, $SD = .86$) than participants in the "informative style" condition: $M = 5.70$, $SD = 1.31$, F (1,113) $= 3.81$, $p = .05$, $\eta_p^2 = .03$. This demonstrates that "persuasive" message was seen as more persuasive than the "informative" message.

3.2.2. Main analysis

A 2 (perceived purpose: to inform vs. to persuade) $\times$ 2 (message style: informative vs. persuasive) ANOVA was conducted on the measure of trust in scientists. Neither of the main effects was significant (Fs < 1.23, ps $> .27$), however there was a significant interaction between perceived purpose and message style: F (1,111) $= 15.88$, $p < .001$, $\eta_p^2 = .13$, see Fig. 1. Pairwise comparisons revealed that participants who were led to believe that scientists' goal is to inform, responded to the informative message with higher trust ($M = 5.32$, $SD = .84$) than to the persuasive message ($M = 4.49$, $SD = .81$; F(1,111) $= 10.95$, $p = .001$, $\eta_p^2 = .09$). In contrast, participants who were led to believe that scientists' goal was to persuade the public to take action, reported higher trust in response to the persuasive message ($M = 4.99$, $SD = .95$) than to the informative message ($M = 4.43$, $SD = 1.08$; F(1,111) $= 5.31$, $p = .023$, $\eta_p^2 = .05$). Said differently, participants who were exposed to an informative message, responded with higher trust when they expected to be informed rather than persuaded: F (1,111) $= 13.41$, $p < .001$, $\eta_p^2 = .11$, while participants who received a persuasive message reported higher trust when expected to be persuaded rather than informed: F (1,111) $= 4.00$, $p = .048$, $\eta_p^2 = .04$.

The same 2 $\times$ 2 ANOVA was conducted on the measure of environmental concern. Again, neither of the main effects was significant (Fs < 2.30, ps $> .13$), but there was a significant interaction between perceived purpose and message style: F (1,111) $= 5.39$, $p = .022$, $\eta_p^2 = .05$, see Fig. 2. Pairwise comparisons demonstrated that participants who expected to be informed responded to the informative message with stronger concern ($M = 6.05$, $SD = 1.00$) than to the persuasive message ($M = 5.58$, $SD = 1.27$), although this difference did not reach statistical significance: F (1,111) $= 2.60$, $p = .110$, $\eta_p^2 = .02$. In contrast, participants who expected to be persuaded, reported marginally stronger concern in response to persuasive ($M = 5.74$, $SD = .83$) rather than informative message: $M = 5.28$, $SD = 1.24$, F (1,111) $= 2.81$, $p = .097$, $\eta_p^2 = .03$. Said differently, participants who received the informative message were more concerned by it when they expected to be informed rather than persuaded: F (1,111) $= 7.62$, $p = .007$, $\eta_p^2 = .06$. Among participants who received the persuasive message there were no significant differences according to expectations: F (1,111) $= .31$, $p = .577$, $\eta_p^2 < .01$.

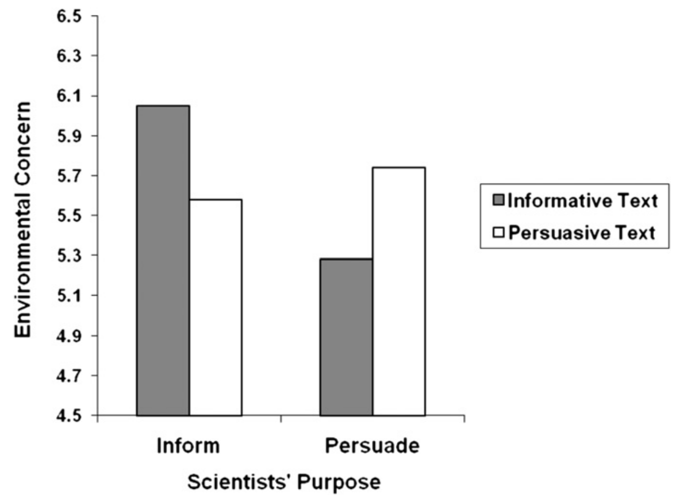

Fig. 2. Mean level of environmental concern as a function of perceived communicative purpose and message style.

A similar pattern of results was obtained on the measure of environmental intentions. Again, neither main effect was significant (Fs < 1.09, ps $> .29$), but there was a significant interaction between the perceived goal of scientists and message style: F (1,111) $= 7.46$, $p = .007$, $\eta_p^2 = .06$, see Fig. 3. Pairwise comparisons showed that participants who believed that scientists' goal was to inform, reported stronger sustainable intentions in response to informative ($M = 3.97$, $SD = 1.36$) rather than persuasive message ($M = 3.14$, $SD = 1.10$), F(1,111) $= 6.21$, $p = .014$, $\eta_p^2 = .05$. Intentions of those who believed that scientists' goal was to persuade were not significantly affected by the message style: $M_{\text{persuasive}} = 3.85$, $SD = 1.28$, $M_{\text{informative}} = 3.42$, $SD = 1.10$, F (1,111) $= 1.82$, $p = .180$,

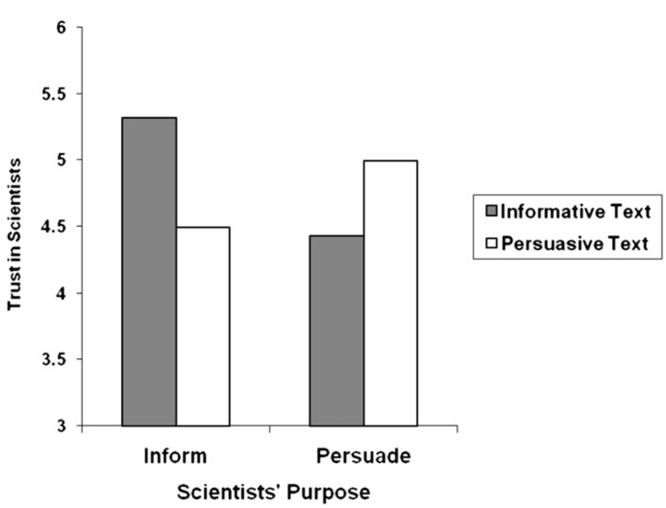

Fig. 1. Mean level of trust in climate scientists as a function of perceived communicative purpose and message style.

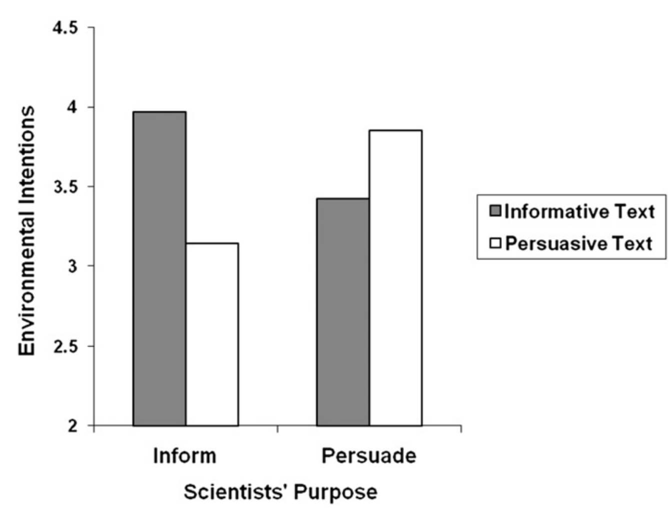

Fig. 3. Mean level of environmental intentions as a function of perceived communicative purpose and message style.

$\eta_p^2 = .02$. Said differently, participants who received the informative message, reported marginally stronger sustainable intentions when they expected to be informed rather than persuaded: $F(1,111) = 2.92$, $p = .090$, $\eta_p^2 = .03$, while participants who received the persuasive message responded with marginally stronger intentions when they expected to be persuaded rather than simply informed: $F(1,111) = 3.77$, $p = .055$, $\eta_p^2 = .03$.

3.2.3. Mediation analysis

Given the parallel effects of perceived communicator's purpose and message style on environmental concern, intentions and trust in scientists, mediation via trust was a possibility. To further establish this, we conducted a mediated moderation analysis via regression following the procedures described by Muller, Judd, and Yzerbyt (2005). The results of this analysis met the conditions for mediation via trust for both environmental concern and intentions as dependent variables. Briefly, in addition to the significant interaction between perceived scientists' purpose and message style on environmental concern ($\beta = .38$, $p = .022$), and trust ($\beta = .63$, $p < .001$), trust was itself a significant predictor of concern ($\beta = .48$, $p < .001$). In addition, when trust was included as a predictor of concern, the previously significant interaction between the perceived goal of scientists and message style became non-significant ($\beta = .09$, $p = .562$), and trust remained a significant predictor of concern ($\beta = .40$, $p = .004$).[1] Using a bootstrapping analysis (Preacher & Hayes, 2004), the bias corrected bootstrap estimate of the indirect effect of the interaction between perceived scientists' purpose and message style had a 95% confidence interval of .2687 to 1.2823. As this confidence interval does not cross zero, this suggests a significant pattern of mediation on environmental concern via trust.

The same mediation pattern was evident with environmental intentions as a dependent variable. In addition to the significant interaction between perceived scientists' purpose and message style on intentions ($\beta = .44$, $p = .007$), trust was a significant predictor of intentions ($\beta = .33$, $p < .001$). When trust was included as a predictor of intentions, the interaction between perceived goal and message style became non-significant ($\beta = .27$, $p = .113$), and trust remained a significant predictor of intentions ($\beta = .29$, $p = .047$).[2] The bias corrected bootstrap estimate of the indirect effect had a 95% confidence interval of .1668 to .9938. Again, this confidence interval does not include zero and this suggests a significant pattern of mediation on environmental intentions via trust in climate scientists.

3.3. Discussion

The results of Study 2 support our suggestion that participants' responses to different types of scientific messages are affected by expectations about scientists' communicative goals. The more specific pattern was somewhat different from what might have been predicted based on previous research that has investigated the effects of prior expectations on responses to communication. In the context of scientific communication, confirmed expectations seemed to motivate trust and responsiveness more than disconfirmed expectations (cf. Terwel et al., 2009). Specifically, participants who expected to be informed, responded to informative messages with higher trust and stronger sustainable intentions than to persuasive messages. Similarly, participants who expected to be persuaded, responded to *persuasive* messages with higher trust and stronger intentions than to informative messages. Importantly, this increase in trust as a function of fit between communicator goal and message translated into higher willingness to act in line with the message (i.e., responsiveness).

4. General discussion

This paper aimed to explore how the perceived goals of science communicators might shape public responses to science communication. Based on previous research on the impact of communicators' and recipients' expectations about each other on the communication process, it was expected that the public's expectations regarding scientists' communicative motives would have a significant impact on responses to climate science messages. This hypothesis received empirical support in two experimental studies.

Study 1 demonstrated that the degree to which participants were willing to trust climate scientists and engage with the issue of climate change depended on their perception of scientists' communicative purpose—a perception that scientists were motivated by impartiality elicited more trust than the perception of a persuasive agenda. Study 2 considered how the style of a communicated message might interact with these prior perceptions to shape receptiveness to the message. The results suggest that fit between participants' understanding of scientists' motives and the style of scientific messages produced higher trust and stronger sustainable intentions. Participants who believed that scientists are motivated to provide impartial information responded with higher trust to purely informative messages. In contrast, participants who were led to believe that scientists' purpose is to persuade the public to take action were more trusting of messages constructed in a persuasive style. Importantly, in both cases the increased trust translated into stronger willingness to take the climate change message on board and to engage in sustainable action. This somewhat qualifies the impression that persuasive goals in science might be harmful for public communication. Rather, consistency between motives and messages seems more important for trust and responsiveness than the pursuit of any particular goal. The theoretical and practical implications of these findings are considered below.

4.1. Theoretical implications

These results are broadly consistent with previous research on the role of communication partners' perceived motives (e.g., Giles et al., 1991; Eagly, Chaiken, & Wood, 1981). In line with this previous research, our results suggest that perceptions of a communicator's motives can alter willingness to engage with their message. Also consistent with previous findings (e.g., Siegrist, 2000), our results demonstrate that the process of trust development is crucial to understanding these effects: differences in willingness to act in line with the message were mediated by trust in source in both studies.

At the same time, some of our specific findings seem to be inconsistent with previous research, and the initial hypotheses that were derived from this work. In particular, on the basis of research by Eagly and colleagues (Eagly et al., 1981) it could have been expected that receiving a purely informative message when one expects to be persuaded would result in higher rather than lower trust in communicator. According to this approach, this kind of discrepancy between perceived motive and ultimate message should signal that a communicator has overcome his or her bias, resulting in the impression that they are actually trustworthy. In contrast to these predictions (and in line with research by Terwel et al., 2009), our results suggest that participants are more

[1] Neither trust by message style, nor trust by perceived motive interactions were significant ($\beta = .07$, $p = .614$ and $\beta = .17$, $p = .140$ respectively).

[2] Neither trust by message style, nor trust by perceived motive interactions were significant ($\beta = -.01$, $p = .946$ and $\beta = .12$, $p = .318$ respectively).

trusting of messages that fit their expectations, even when these expectations concern communicators' persuasive motives. Apparently, in the present context it seems more important for communicators to demonstrate consistency than impartiality.

To reconcile these diverging patterns, it is important to note that previous research focused on the mis-fit between recipients' expectations regarding the *content* of the message and the actual content. In contrast, in the present research we explored (in) consistency between expectations about the *style* of the message (informative versus persuasive) and the actual message style. It is possible to suggest that content inconsistency leads to increased trust (due to attributing change of opinion to communicator's honesty), while style inconsistency, in contrast, undermines trust (perhaps, due to searching for signs of persuasion even in purely informative messages). Future research could further explore these two types of inconsistency.

Our findings seem also inconsistent with previous research on resistance to persuasion (e.g., Jacks & Devine, 2000). According to this previous work, recipients who expect to be persuaded should respond by decreasing their trust in the source and actively resisting the message. Although the results of Study 1 are in line with this suggestion, the results of Study 2 suggest that expectations to be persuaded do not necessarily have fatal consequences for communication outcomes. In fact, participants whose expectations were confirmed by receiving a persuasive message reported the same level of trust and receptiveness to the message as participants who received an informative message (which confirmed their expectations to be informed). Unlike previous research on resistance to persuasion, our studies were conducted in the domain of science communication. Since scientists are seen as a highly credible source of information, it is possible that persuasive attempts coming from scientists are seen as legitimate and trigger little resistance. This explanation is consistent with the evidence that when message recipients are not motivated to counter-argue, they are less sensitive to warnings of impending persuasive attempts (Chen, Reardon, Rea, & Moore, 1992; Romero, Agnew, & Insko, 1996; Wood & Quinn, 2003). Future research could explore the role of perceived legitimacy of persuasive attempts in science communication, as well as the impact of other assumptions about the nature of science (for example, whether the science is seen as a tool for changing the world or acquiring pure theoretical knowledge).

4.2. Practical implications

The present findings have a number of practical implications for communicating science in general, and climate science in particular. Specifically, rather than suggesting that science communicators should only ever portray themselves as agents of information rather than influence, our results suggest that actively managing audience's expectations might be the key to effective science communication. A wide-spread perception of scientists as a credible source of information may be relied on in order to create specific expectations about the style of communication between scientists and the general public (i.e. the public may be open to scientists' attempts to explicitly set the rules of communication). The minimal nature of our experimental manipulations suggests that this may be achieved relatively easily. However, the data suggest that forming expectations may not be sufficient for successful communication; it is equally important to follow them by consistently framed messages. For example, if the public expects the science communication process to be purely informative, it is important for the successful trust building that scientific messages avoid persuasive angle. At the same time, if scientists aim to deliver a persuasive message, they may need to manage public expectations accordingly. As our results demonstrate, a persuasive agenda is not always damaging for trust. If followed by messages constructed in a persuasive style, it can lead to increased public engagement. Overall, it seems that effective expectations management and consistency are vital elements of successful communication between the science world and the general public.

4.3. Limitations and directions for future research

It must be pointed out that the message style manipulation used in Study 2 was relatively subtle. In particular, in an attempt to make the two messages as similar as possible in terms of their other characteristics, we used a mild persuasive message (avoiding openly persuasive rhetoric). On one hand, the subtlety of the manipulation is a strength of the present study, since it demonstrates that people are sensitive to even minimal differences in message style. On the other hand, this leaves open the question of how people would respond to messages created in a more openly persuasive style (which is often the case with real world persuasive messages). At the same time, it should be noted that the style of both messages was evaluated by participants as relatively persuasive (judging by the means on the manipulation check measure). This may suggest that the effects might be even stronger when the messages are seen as devoid of any persuasive intent. Future research could explore whether the observed effects hold (and whether they are strengthened) for less subtle manipulations of message style.

Another direction for future research is exploring how people develop an understanding of scientists' motives. Although in the present set of studies we manipulated the perceived motives directly, it would be important to understand what factors affect these beliefs about science communicators. For example, future research could explore such factors as level of education, environmental concern or general perceptions of the role of science in society as determinants of perceived scientists' motives.

Similarly, it would be important to explore individual differences that might moderate the present findings. In particular, the present samples are biased towards female participants. Since there is evidence of gender differences in risk tolerance and trust (see Swim et al., 2011), future research may benefit from exploring the effects of perceived scientists' motives on responses to science communication in more gender-balanced samples. In addition, it might be fruitful to look at the role of environmental concern on the observed effects. Although we would predict that environmental concern would have a main effect on responsiveness to climate change messages, it could also be possible that those high on environmental concern are more sensitive to consistency in climate change communications. Future research could investigate these alternative possibilities.

4.4. Conclusion

This paper explored the effect of recipients' expectations about communicators' motives on trust in communicators and responsiveness to different types of scientific messages (informative versus persuasive). The results revealed that consistency between expectations and message style led to higher trust in message source. Specifically, participants who expected to be persuaded responded with higher trust to persuasive (rather than purely informative) messages, while the opposite was true for those who expected to be informed. Importantly, in both cases higher trust led to stronger willingness to behave in line with the message. The results qualify previous findings on resistance to persuasion by demonstrating that expectations to be persuaded can be effectively managed by stating intentions clearly and delivering messages consistent with these expectations. The findings have important implications for communicating science: they suggest that openness about the purpose of communication and consistency between the stated aims and the actual behaviour are key to developing trust and understanding between the scientific world and the public.

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